

100 Days of Linux - Day 041

Title: Creating and Extracting Archives with tar

Today's Goal

Learn how to create, list, and extract TAR archives using the **tar** command.

Why This Matters

Linux professionals use TAR archives to back up projects, move datasets, and package source code before compression or deployment.

Command of the Day: tar

Common Syntax:

```
tar -cvf archive.tar folder/
```

```
tar -xvf archive.tar
```

```
tar -tvf archive.tar
```

Options:

-c = Create archive

-x = Extract archive

-t = List archive contents

-v = Verbose output

-f = Archive filename

Example:

```
tar -cvf project_backup.tar retail_project/
```

Beginner Lesson

A TAR archive bundles many files and folders into a single file without compressing them. Compression will be covered in the next lessons.

Practice Questions

1. Create a folder named `project_backup` containing `README.md`, `employees.csv`, and `application.log`. Create a TAR archive named `project_backup.tar`.
2. List the contents of `project_backup.tar` without extracting it.
3. Create a folder named `restore` and extract `project_backup.tar` into it.
4. Verify that all extracted files exist using `ls`.
5. Display your current working directory after completing today's tasks.

Hints

Remember: create (-c), list (-t), extract (-x). Always include -f before the archive name.

Mini Challenge

Create a `data_pipeline` folder with `data`, `logs`, and `scripts` subfolders. Archive it, list the contents, extract it into another folder, and compare both structures.

Common Mistakes

Confusing TAR archives with compressed files. A .tar file is archived but not compressed.

Did You Know?

The name TAR comes from 'Tape Archive' because it was originally designed for magnetic tape backups.

Pro Tip

Use tar to package an entire project before copying it to another machine or uploading it.

Answer Key (Instructor Only)

Q1: tar -cvf project_backup.tar project_backup/

Q2: tar -tvf project_backup.tar

Q3: mkdir restore && tar -xvf project_backup.tar -C restore/

Q4: ls restore/project_backup

Q5: pwd

Homework

Create three different project folders and practice creating, listing, and extracting TAR archives until you're comfortable with the workflow.

Next Day Preview

Tomorrow you'll learn gzip to compress files and combine it with tar to create .tar.gz archives.